

PRODUCT DATA EXPLORATION USING MICROSOFT EXCEL

Introduction

This project focuses on analyzing product-related data using Microsoft Excel. The dataset includes important details such as Product ID, Product Name, Brand Name, Category, Price, and Quantity. Excel formulas and spreadsheet functions are used to organize and analyze the data effectively.

The project helps in understanding product information, pricing, quantity management, and spreadsheet-based data analysis.

Objective

The main objective of this project is to perform product data analysis using Microsoft Excel and improve practical knowledge in spreadsheet management and formula implementation.

Software Requirements

The following requirements are needed to execute this project:

- Microsoft Excel 2016 or later
- Windows 10 or above
- Basic knowledge of Excel formulas and tables

Dataset Description

The dataset contains product-related information used for analysis, including:

- Product ID
- Product Name
- Brand Name
- Category
- Price
- Quantity

This information helps in understanding product distribution, pricing, and quantity details.

Methodology

The project was completed using the following steps:

- Opened the Excel workbook.
- Reviewed the product dataset.
- Applied Excel formulas for calculations.
- Analyzed category-wise products, pricing, and quantity information.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	Standard price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard price	20	CN	AUG
27-JAN-IT	Laptop	Acer	980	10	Electronics	High price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High price	29	CA	SEP
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard price	03	CA	JUN
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard price	11	CA	JUL
16-APR-ES	Headphones	Bose	230	20	Electronics	Standard price	16	ES	APR
07-MAR-CA	Dress	Zara	60	30	Fashion	Standard price	07	CA	MAR
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard price	13	CA	APR
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard price	24	CA	MAY
02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard price	02	CA	DEC
17-JUN-IN	Laptop	HP	950	25	Electronics	High price	17	IN	JUN
09-JUL-FR	Watch	Casio	100	20	Accessories	Standard price	09	FR	JUL

TASK 1	Sum, Count, Average
Total price of all product	10100
How many product exist in the data base	34
Average price of the product	297.0588
TASK 2	Minimum, Maximum values
Minimum price	30
Maximum price	1000
TASK 3	Logical Function -IF
Create a new column named Price Range	
TASK 4	Conditional Function - SUMIF & COUNTIF
Total price of products in the electronic category	8050
Number of product with a price less than 100	11
TASK 5	Text formatting - LEFT, RIGHT, MID
Created a 3 column named Day, Country code, Month	

Excel Formulas Used

SUM Formula

Used to calculate the total value.

=SUM(D2:D34)

AVERAGE Formula

Used to calculate the average value.

=AVERAGE(D2:D34)

COUNTA Formula

Used to count non-empty cells.

=COUNTA(B2:B34)

MAX Formula

Used to identify the highest value.

=MAX(D2:D34)

MIN Formula

Used to identify the lowest value.

=MIN(D2:D34)

Learning Outcome

This project improved practical knowledge in Microsoft Excel, spreadsheet organization, formula implementation, and product data analysis. It also enhanced problem-solving and data-handling skills.

Conclusion

This project demonstrates how Microsoft Excel can be effectively used for product data analysis. By applying formulas and spreadsheet techniques, product-related information was successfully organized and analyzed.

References

Microsoft Excel

GitHub Repository

Online Excel Learning Resources